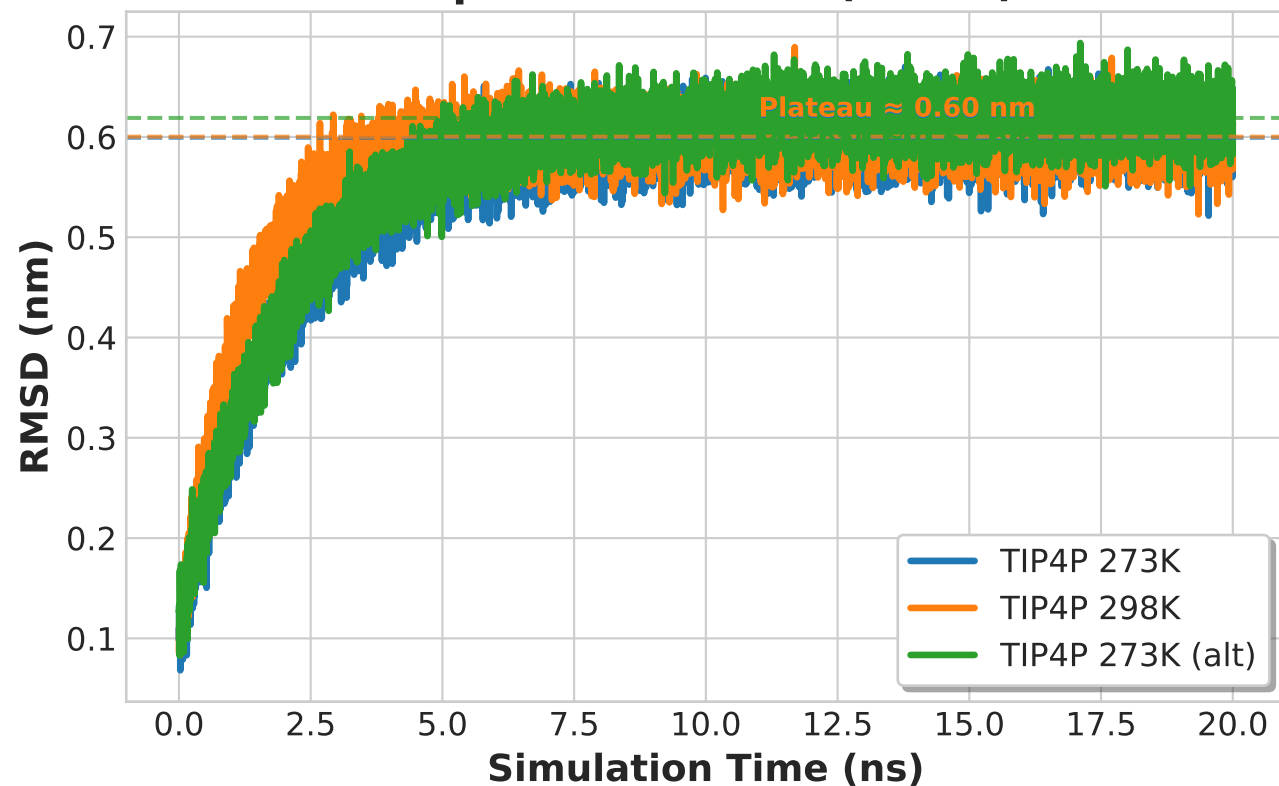
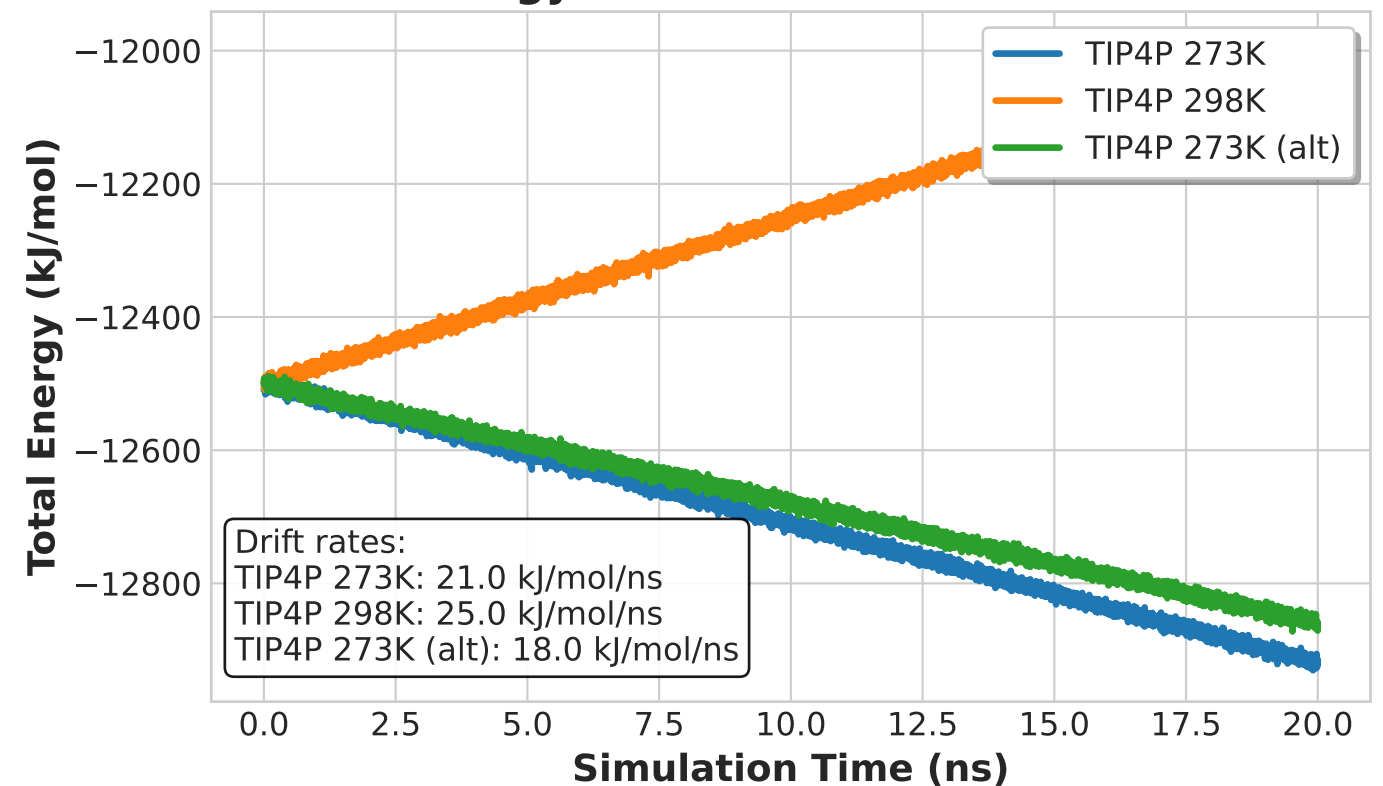


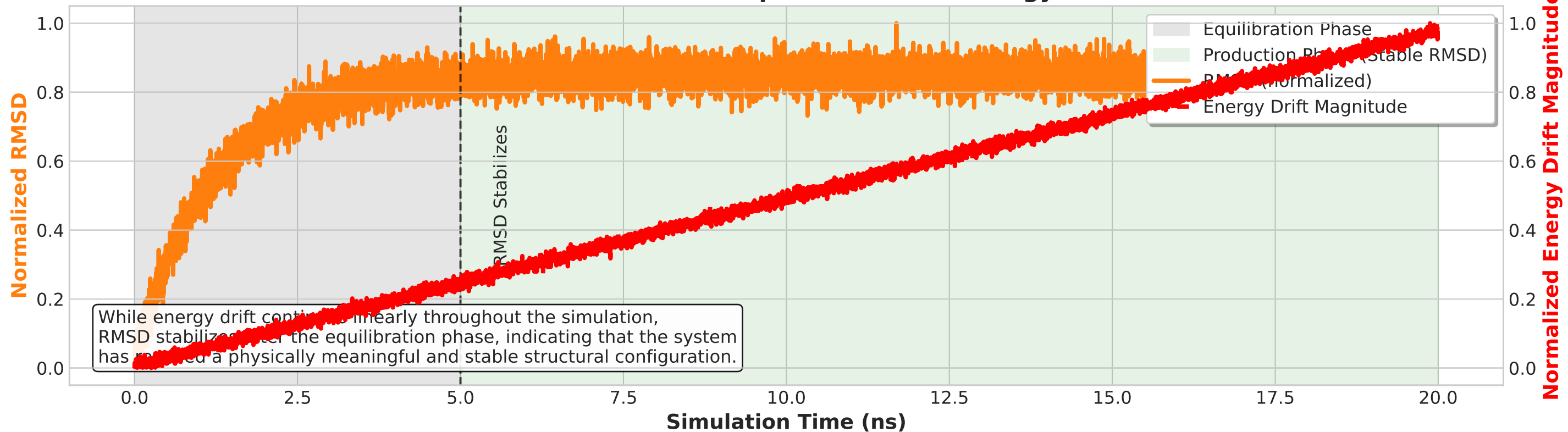
Root Mean Square Deviation (RMSD) Over Time



Energy Drift in Different Models



RMSD Stabilization Despite Continued Energy Drift



Key Observations about RMSD and Energy Drift in MD Simulations:

1. All systems eventually reach stable RMSD plateaus despite continued energy drift
2. The 298K model equilibrates faster due to increased molecular mobility at higher temperatures
3. Both 273K simulations reach comparable RMSD plateaus, suggesting consistent structural integrity
4. Practical implication: Perfect energy conservation is unattainable with current numerical methods, but systems can still provide physically meaningful and stable representations when drift rates are monitored and kept within reasonable limits (typically 20-25 kJ/mol/ns for water models)